

justifications contain hallucinated novelty-aspect arguments (i.e., those unsupported by the corresponding research idea). Using the same approach as in *Known Aspects Hallucination Rate*, we compute this metric with Equation 3, where $N_{\text{hallucinated}}$ represents the number of model-generated novelty-aspect arguments unsupported by the research idea and $N_{\text{generated}}$ is the total number of novelty-aspect arguments in the model-generated justification.

For all LLM-based evaluations, it is crucial to use a high-performing model to ensure the accuracy of the various metrics. Accordingly, we use the GPT-4.1 (OpenAI, 2025b) model for all LLM-based evaluations. Additionally, each presented justification metric is computed per individual sample. To provide a comprehensive evaluation in RINoBench, we average the computed justification metric scores across multiple samples.

5. Benchmarking LLMs as Judges of Research Idea Novelty

In this section, we present a benchmarking study examining several state-of-the-art LLMs on their ability to judge the novelty of research ideas. To this end, we follow recent works that frame the novelty judgment of research ideas as zero-shot task by directly giving the review criteria and prompting LLMs for a final score (Yang et al., 2024; Lu et al., 2024; Si et al., 2025; Li et al., 2025b; Baek et al., 2025). Accordingly, we instruct each LLM to perform the RINoBench task illustrated in Figure 1 to generate a numerical novelty score and a textual justification. Thereby, the LLM is provided with the novelty judgment rubric detailed in Table 1, alongside a research idea and its related works. The LLM is then asked to analyze the research idea, identify its key contributions, and compare it to the provided related works. Based on this analysis and comparison, the model is finally tasked to generate a suitable novelty score according to the rubric, accompanied by a brief justification explaining its reasoning for the predicted score. The exact instructions are shown in Figure 6.

For this study, we select a diverse set of LLMs, encompassing a range of sizes and reasoning capabilities. Specifically, we include non-reasoning models including *Llama-3.1-8B* (Grattafiori et al., 2024), *Llama-3.3-70B* (Grattafiori et al., 2024), and *Llama-4-Scout-17B-16E* (Meta, 2025), as well as reasoning-capable models including *DeepSeek-R1* (DeepSeek-AI et al., 2025), *GPT-OSS-120B* (OpenAI et al., 2025), *o3* (OpenAI, 2025c), and *GPT-5* (OpenAI, 2025a). This selection enables a comprehensive evaluation of model performance across different architectures and reasoning abilities.

Novelty Judgment Performance Table 3 shows the evaluation results. We observe that all models show very low novelty judgment abilities, with none of them achieving significant F_1 scores and the highest macro-average being 17.2. Notably, no model successfully predicted the novelty category 1, as indicated by the 0.0 F_1 scores for this category across all models. This suggests a strong bias against predicting ideas as "not novel", indicating that the models tend to avoid judging ideas as lacking novelty altogether. Interestingly, novelty categories 2 and 5 are occasionally predicted correctly, but the models predominantly predicted novelty categories 3 and 4. This suggests that the models tend to avoid assigning extreme values of novelty (such as "marginally novel" or "highly novel and innovative"). Instead, they consistently attempt to find at least some aspect of novelty in a research idea, even if it is not present. The MAE values are relatively low and consistently hover around 1, indicating that while the models may often make errors in their novelty judgments, these do not deviate drastically from the gold standard.

Quality and Coverage of Justifications The justification metrics provide interesting insights into how the models substantiate their novelty judgments. It is noteworthy that all models exhibit relatively high recall, implying that the LLMs frequently use arguments similar to those found in the human-annotated gold standard justifications. This finding is consistent with the results of Afzal et al. (2026), who reported high alignment between LLM and human novelty reasoning. Moreover, the high additional ratios (Add. Ratio), often exceeding 100%, suggest that the LLMs tend to generate more elaborate justifications than humans and frequently find more arguments to justify their novelty predictions. This may be because, for humans, one or two well-chosen arguments are often sufficient to judge the novelty of a research idea, while LLMs appear to strive to provide a comprehensive set of arguments, likely in an attempt to satisfy the user.

Hallucinations and Judgment–Justification Gap

Despite generating many arguments, the hallucination rate is low across all models, suggesting that the models' justifications are largely grounded in the provided context. This indicates that, while the models may sometimes over-elaborate in their novelty judgment justifications, the arguments they generate are mostly reliable and supported by evidence. This stands in contrast to their novelty score predictions, which are more dissimilar from the human-annotated gold novelty scores, pointing to a gap in the models' ability to accurately judge the novelty of research ideas, even if their justifications seem to contain plausible arguments.

Model	Novelty Score Metrics							Justification Metrics						
	F ₁						MAE	ALI	Recall		Add. Ratio		Hall. Rate	
	Macro	1	2	3	4	5			KA	NA	KA	NA	KA	NA
Non-Reasoning Models														
Llama-3.1-8B	14.6	0.0	0.0	26.2	41.3	5.4	1.00	0.58	85.5	75.3	62.8	111.6	4.2	3.4
Llama-3.3-70B	9.5	0.0	0.0	2.2	45.0	0.0	1.04	0.55	88.9	78.3	86.3	115.3	1.1	1.4
Llama-4-Scout	13.0	0.0	0.0	17.1	42.7	5.1	1.01	0.58	89.8	81.9	89.0	120.3	0.0	1.1
Reasoning Models														
GPT-OSS-120B	14.6	0.0	3.0	30.1	40.7	0.0	0.96	0.64	88.1	77.8	79.0	92.4	0.9	0.5
DeepSeek-R1	12.3	0.0	0.0	16.1	45.6	0.0	0.99	0.67	87.8	81.1	115.7	112.4	0.6	0.2
o3	16.2	0.0	5.6	35.6	39.7	0.0	0.93	0.72	90.4	85.6	139.9	74.0	1.3	1.7
GPT-5	17.2	0.0	16.7	32.2	37.3	0.0	0.93	0.71	89.9	85.7	122.1	91.8	0.6	0.5

Table 3: Evaluation results of novelty judgments on the RINoBench test set. As described in Section 4.2.1, the reported novelty score metrics include F_1 macro averaged and for each rubric category (1-5), as well as MAE. Further, as outlined in Section 4.2.2, the justification metrics include *Alignment (ALI)*, as well as *Recall*, *Additional Ratio* (in %), and *Hallucination Rate* (in %) for *Known Aspects (KA)* and *Novelty Aspects (NA)* respectively.

Reasoning vs. Non-Reasoning Models When comparing model performance, we observe that reasoning models outperform their non-reasoning counterparts, albeit by a small margin. The GPT-5 model achieves the highest macro-averaged F_1 score with 17.2, closely followed by o3 (16.2). These models, designed for more complex reasoning tasks, outperform the non-reasoning models, which generally exhibit lower F_1 scores and demonstrate worse novelty judgment abilities. This indicates that incorporating additional reasoning and deeper thinking during the generation process helps models make more accurate judgments regarding the novelty of research ideas.

Takeaway Overall, the results show that the LLM-generated novelty judgment justifications closely align with those of human experts, whereas the predicted novelty judgment scores diverge substantially from the human-annotated gold scores, pointing to a clear gap: *although LLMs can generate plausible and well-supported novelty justifications, they fail to translate their reasoning into accurate novelty judgments*. Further, the models tend to avoid extreme predictions and avoid judging ideas as *not novel at all* nor *highly novel and innovative*. Instead, they constantly strive to find a middle ground. Additionally, while the models’ justifications are often grounded in the provided context, they tend to be more elaborate than human justifications, reflecting a difference in how humans and models approach novelty judgment. Despite these differences, the models’ predictions are only slightly different from human annotations (see MAE scores), suggesting that their beliefs about novelty do not differ fundamentally, but are instead somewhat inaccurate or imprecise.

6. Conclusion

This work introduces RINoBench, the first automated benchmark for evaluating novelty judgments of research ideas. It includes 1,381 research ideas derived from and judged by human experts. Further, the benchmark comprises nine automatic metrics to assess the accuracy of predicted novelty scores and to compare model-generated textual justifications with human-authored gold justifications. Our work bridges the gap between current, largely manual and incomparable human evaluations, towards reproducible and comparable evaluations.

Further, we investigate the capability of several state-of-the-art LLMs to judge the novelty of research ideas. Our findings reveal that current LLMs face substantial challenges in accurately judging the novelty of research ideas. Notably, while LLMs refrain from judging ideas as completely lacking novelty, they tend to seek a middle ground, aiming to attribute at least some degree of novelty while simultaneously avoiding judgments of high novelty and innovation. Interestingly, as indicated by the strong correspondence between LLM-generated and human-authored justifications, LLM reasonings align closely with human rationales for research idea judgment. However, this alignment does not translate to the LLM-predicted novelty scores, which diverge considerably from human-assigned scores. Finally, while all LLMs examined exhibit difficulties in effective novelty judgment, our experiments indicate that reasoning-capable models consistently outperform non-reasoning ones, suggesting that longer thinking and deeper engagement with the input and task instructions improve LLM-based novelty judgment of research ideas.

7. Limitations

RINoBench is derived exclusively from ICLR 2022 and 2023 submissions, limiting its domain, epistemic, and cultural diversity. Because it reflects a single conference ecosystem centered on machine learning, the benchmark captures the reviewing norms and novelty criteria of that community. Fields with different epistemological assumptions and evaluation practices—such as many areas in the humanities and social sciences—are not represented. Consequently, the novelty dimensions emphasized in RINoBench, focused on technical or methodological innovation, may underrepresent theoretical or discovery-oriented contributions. Findings based on RINoBench should therefore be interpreted within this specific context and validated on broader, more heterogeneous datasets.

The dataset relies on peer review scores, which are inherently subjective and shaped by disciplinary conventions. While this enables modeling real-world novelty judgments, it also operationalizes a particular reviewing culture rather than a universal notion of novelty, potentially reflecting individual or systemic biases.

Because the data originates from predominantly English-language OpenReview submissions, linguistic and rhetorical conventions may influence how novelty is expressed and assessed. Furthermore, although LLMs are used to extract structured ideas and synthesize justifications with careful filtering, they may introduce hallucinations, normalization effects, or discourse-sensitive distortions.

Finally, RINoBench focuses solely on novelty and does not capture other dimensions of research quality, such as rigor, significance, or reproducibility.

8. Ethics Statement

All data in RINoBench is derived from publicly available OpenReview submissions and papers indexed by Semantic Scholar. No private reviewer or author information is included. We emphasize that the dataset is intended for research and educational purposes only. Users should not use models trained on this data to make formal or high-stakes judgments of research ideas, as novelty judgments are inherently subjective and context-dependent.

9. Broader Impact

RINoBench is designed to advance AI-assisted scientific discovery by enabling models to reason about and explain novel contributions in research. It provides a valuable resource for researchers, educators, and students to understand and teach research idea novelty judgment and serves as a

benchmark for developing models with explainable reasoning capabilities.

At the same time, automated predictions of research idea novelty should not replace human expert judgment. The dataset reflects inherently subjective human opinions and may contain biases from peer reviews. Models developed on RINoBench should be intended as tools to support, rather than replace human judgments of research ideas.

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